21CSC529T Inferential Statistics

Session - 5

Agenda

- Odds Ratio
- Random Variables
- Types of Random Variables
- Probability distributions
- Discrete Probability Distributions
- Continuous probability Distributions
-

Odds

Odds

- Odds of an event is the ratio of the number of observations in favour of an event to the number of observations not in favour of the event

$$\text{odds} = \frac{\text{number of observations in favour of the event}}{\text{number of observations not in favour of the event}}$$

- Probability is sometimes expressed in terms of odds

Odds

Question:

Imagine a bag contains **8 red marbles** and **2 blue marbles**. What is Odds in favor of drawing a red marble ?

Solution : The odds in favor of drawing a red marble are the ratio of the number of red marbles to the number of blue marbles.

Odds in favor of red = 4:1

Odds

What is Odds against drawing a red marble:

Solution : The odds against drawing a red marble are the inverse, representing the chance of drawing a blue marble instead of a red one.

Odds against red = 1:4

Imagine a new diagnostic test is developed for detecting a certain disease. In a clinical trial, the test's results show that the patient having the disease. Result of the test to be positive is 7 and negative is 3.

1) What is the odd in favour of patient to be positive.
2) What is the odd in against of patient to be positive
3) What is the probability of patient to be positive

Answer : 1) Odds in favor of a patient with the disease testing positive: **7:3**

2) Odds against a patient with the disease testing positive: **3:7**

3)Probability of a patient with the disease testing positive: **70%**

Odds Ratio

Imagine we want to check if **smoking is linked to lung disease**.

- Among **smokers**, 60 out of 100 got lung disease.
- Among **non-smokers**, 20 out of 100 got lung disease.

Solution :

Step 1: Calculate odds.

Smokers: 60 diseased / 40 not diseased = **1.5**
Non-smokers: 20 diseased / 80 not diseased = **0.25**

Step 2: Compare with odds ratio.

OR=1.5 / 0.25 = **6**

Interpretation: Smokers have **6 times higher odds** of lung disease compared to non-smokers.

A health researcher wants to study the effect of **regular exercise** on the chances of developing **heart disease**.

The study results for 200 people are summarized below:

Group	Heart Disease	No Heart Disease
Exercise	10	90
No Exercise	30	70

Solution : Calculate odds:

- With Exercise: Odds = 10/90 = 0.11
- Without Exercise: Odds = 30/70 = 0.43

Odds Ratio = 0.11 ÷ 0.43 ≈ 0.26

Interpretation: People who exercise have **lower odds (only about 26%)** of getting heart disease compared to those who don't. Exercise is **protective**.

Summary

Probability of an event is the ratio of number of observations in favour of an event to all possible observations

$$\text{probability} = \frac{\text{number of observations in favour of the event}}{\text{number of observations}}$$

Odds of an event is the ratio of number of observations in favour of an event to number of observations not in favour of the event

$$\text{odds} = \frac{\text{number of observations in favour of the event}}{\text{number of observations not in favour of the event}}$$

Odds

Odds in favour of an event A are a:b

Probability of odd in favour of event A

$$P(A) = \frac{a}{a+b}$$

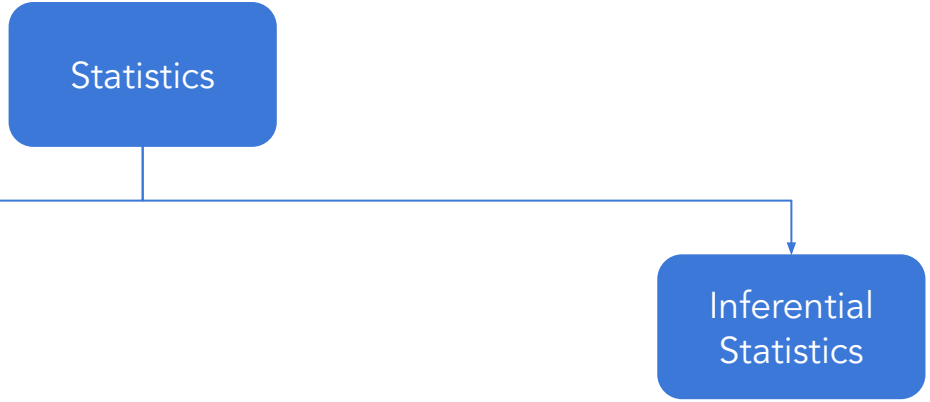
Probability of odd in against of event A

$$P(A') = \frac{b}{a+b}$$

$$P(A') = 1 - P(A)$$

Probability Distributions

Statistics



It is used to only describe the sample or summarize information about the sample

It is used to make inferences and generalizations about the broader population

Random Variable

Random variable

- The action of rolling a die is called a random experiment
- The set of possible outcomes of an experiment is called the Sample Space and it is denoted by Ω
- Possible outcomes for this experiment are: 1, 2, 3, 4, 5, and 6
- A Random Variable is a function defined from the sample space to real numbers

Random variable

- The usual notation used for a random variable is ‘X’. However, other capital letters like U, V, or Z can be used

- Let us observe the experiment of number appeared on the die. We define our random variable as

  X: the number observed

  X takes values 1, 2, 3, 4, 5, and 6.

  Hence $\Omega = \{1, 2, 3, 4, 5, 6\}$

Random variable

Question:

Suppose you are the owner of a grocery shop. You wish to improve the shopping experience of your customers, by easing their payment methods and ensuring that whichever products they demand are available in the shop.

Now collect the data for:

- mode of payment
- bill amount

Can you describe the nature of the data collected?

Random variable

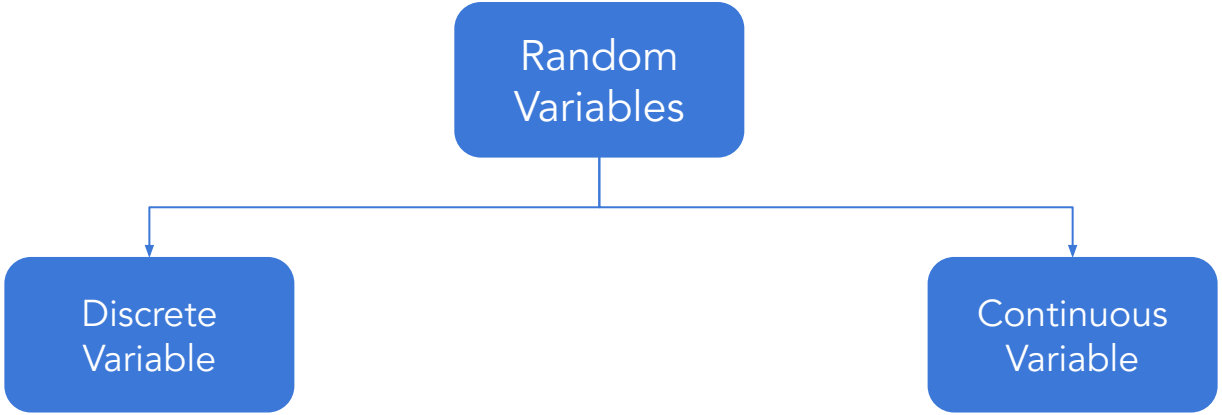
Solution:

The nature of the data collected is as follows:

- mode of payment: the data takes two distinct values *Cashless payment* and *Cash payment*

- bill amount: It takes values such as $11.45, $32, $8.

Random variable



Eg: The mode of payment - cash, mobile wallet, card, bank transfer

Eg: The bill amount

Discrete random variable

- A discrete random variable takes discrete values that is value from the set of whole numbers only or a set of categories
- A discrete random variable can either take finite or countably infinite values
- The possible outcomes can be listed effectively

Discrete random variable

Examples

- Let V : The number of pages in a document
  $\Omega = \{1, 2, 3, ...\}$
- Let X: The number of views for a youtube video
  $\Omega = \{0, 1, 2, 3, ... \}$
- Let U: The size of the shirt available
  $\Omega = \{XS, S, M, L, XL\}$

Continuous random variable

- A continuous random variable takes infinitely many values (within a specified range)
- It can either take an uncountably infinite set of values or values in a given interval

Continuous random variable

Examples:

- Let Z: The income of a person
  Ω = (0, maximum income of a person)

- Let V: The level of water in a damn
  Ω = (0, height of the dam)

Note that random variables Z and V take values from a range and are thus continuous random variables

do it YOURSELF

SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

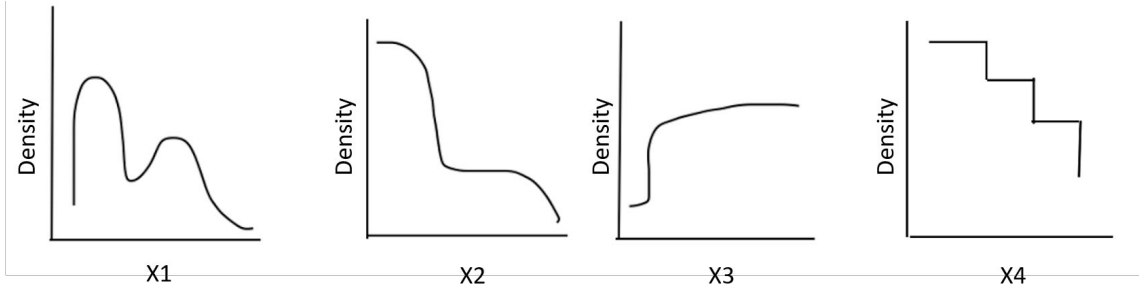
Identify whether the variable is discrete or continuous random variable.

1. Accidents on the New York highway	**Discrete**
2. Fuel capacity of a car	**Continuous**
3. Amount of milk in 15-ounce bottle	**Continuous**
4. Members of the public health society	**Discrete**
5. The total duration of a movie	**Continuous**
6. The number of movies released in a month	**Discrete**

Probability Distributions

Distribution of the data

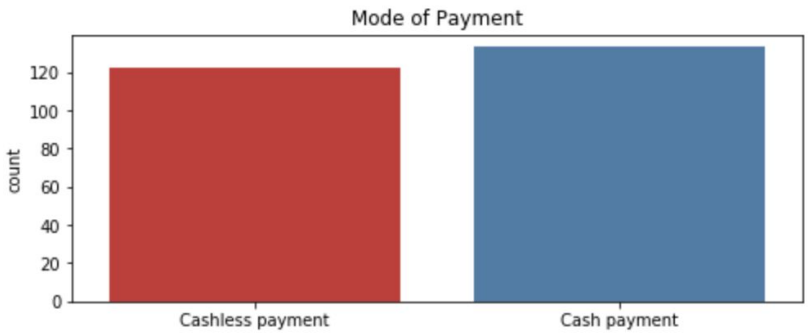
- The distribution is a summary of the frequency of values taken by a random variable
- The distribution of the data gives information on the shape and spread of the data
- On plotting the histogram or a frequency curve for a variable, we actually look at how the data is distributed over its range

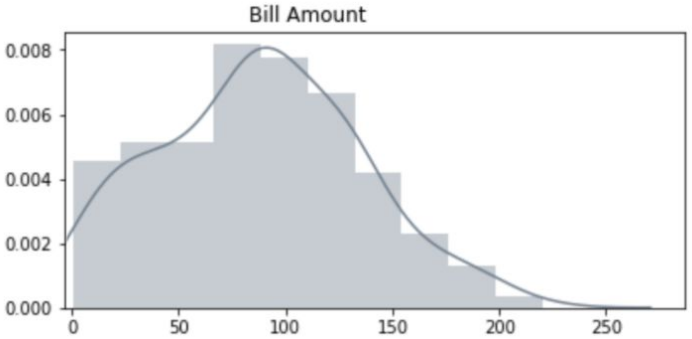


Probability distribution

Recall the example of data collection at the store. The data was collected for the mode of payment and the bill amount.

If we represent the data will have the following figures:





Need for probability distribution

- The statistical tests have assumptions based on the probability distributions that are needed to be verified before using the test
- Once the distribution of the data is known, the descriptive statistics of the data is readily available
- For instance, if the stock trader can determine the probability distribution of a particular stock, then the trader can estimate the potential expected returns that may yield in the future

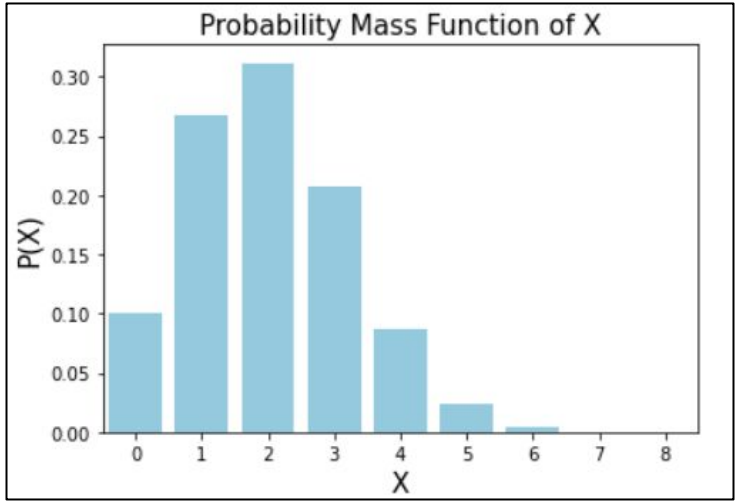
Probability distribution

- A probability distribution is a list of all the possible outcomes of a random variable along with their corresponding probability of occurrence

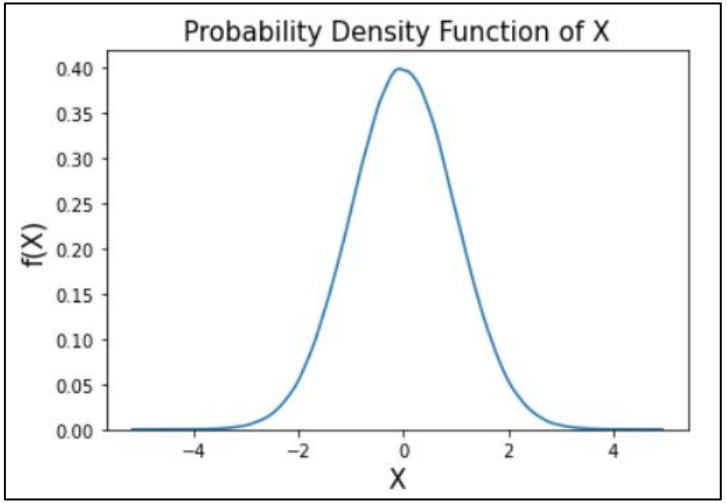
- P(X=x) refers to the probability that the random variable X takes a particular value x

- A probability distribution is defined by its
  - probability mass function (p.m.f) for a discrete random variable
  - probability density function (p.d.f) for a continuous random variable

Probability distribution



Discrete Variable



Continuous Variable

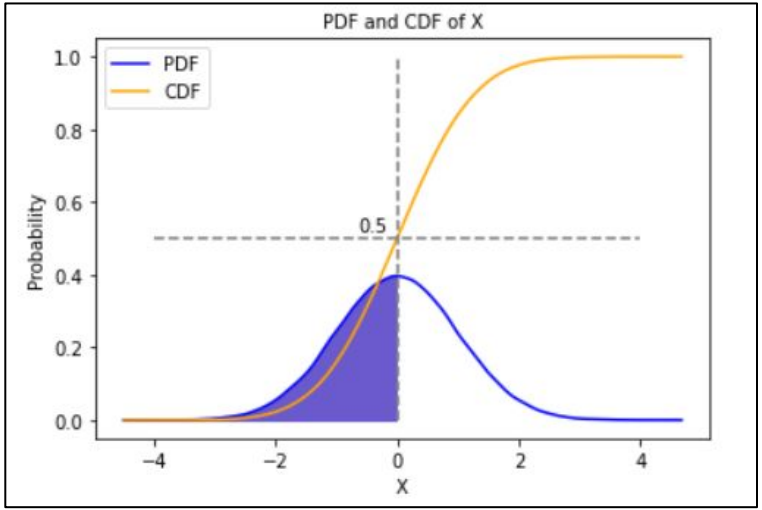
The PMF and PDF difference

The mathematical difference between a discrete distribution and a continuous distribution is:

- The discrete distribution is defined by the probability mass function (pmf) which is the probability at a particular point
- The continuous distribution is defined by the probability density function (pdf) which is the integration over it's range

Cumulative distribution function (c.d.f.)

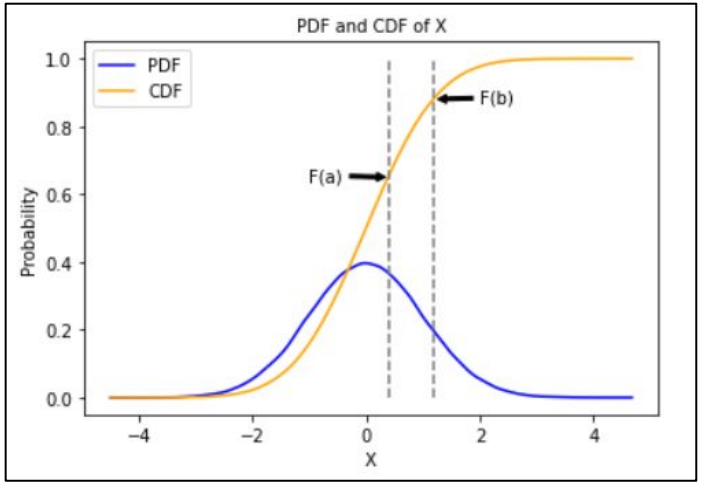
- It gives the cumulative distribution function i.e $P(X \leq x)$
- Denoted by $F(x)$
- That is, $F(x) = P(X \leq x)$
- The area of shaded region is 0.5



Properties of c.d.f.

- $0 \leq F(x) \leq 1$ since it is a probability
- An increasing function or monotonically non-decreasing function
- $F(-\infty) = 0$ and $F(\infty) = 1$
- For any two numbers a and b such that a < b, the probability is computed as:

$$P(a \leq X \leq b) = F(b) - F(a)$$



Mathematical expectation of a random variable

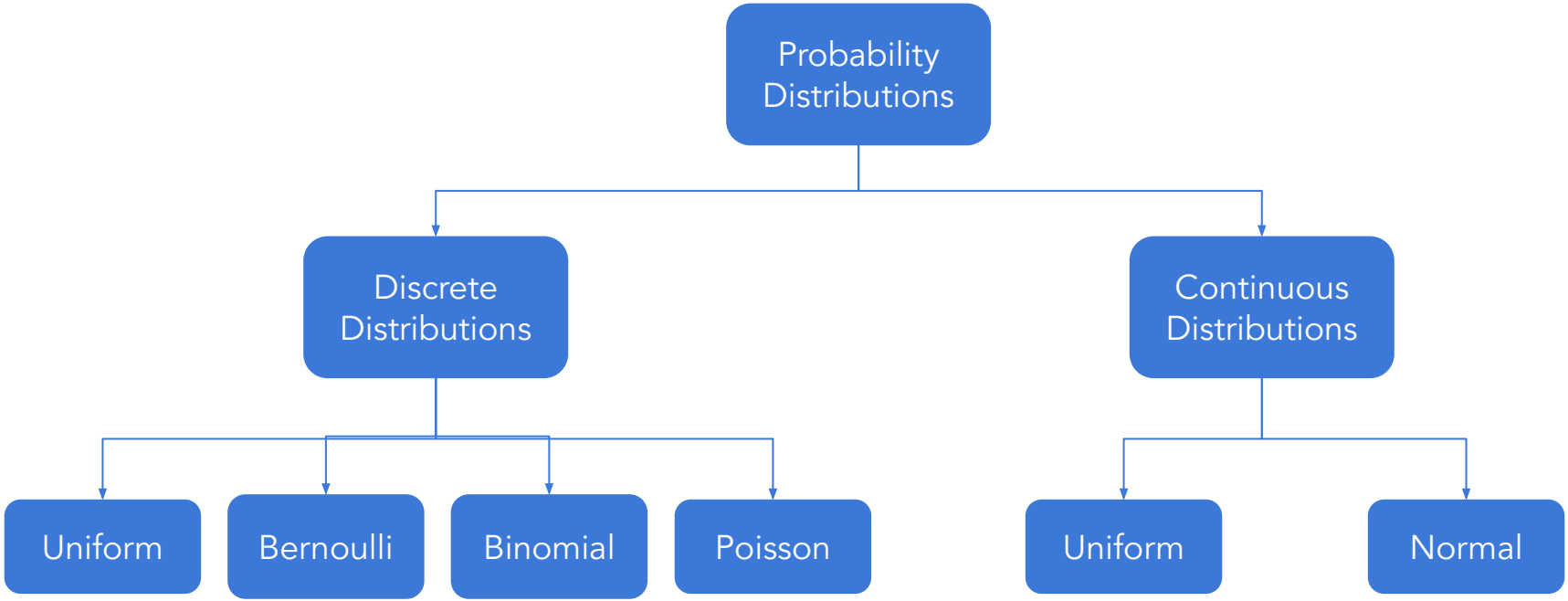
- It is the mathematical mean of a random variable
- It is the central location
- Also known as expected value
- Denoted by μ
- For a probability distribution, it is $\mu = \sum x.\ f(x)$
  Where $f(x)$ is the probability mass function

Variance of a random variable

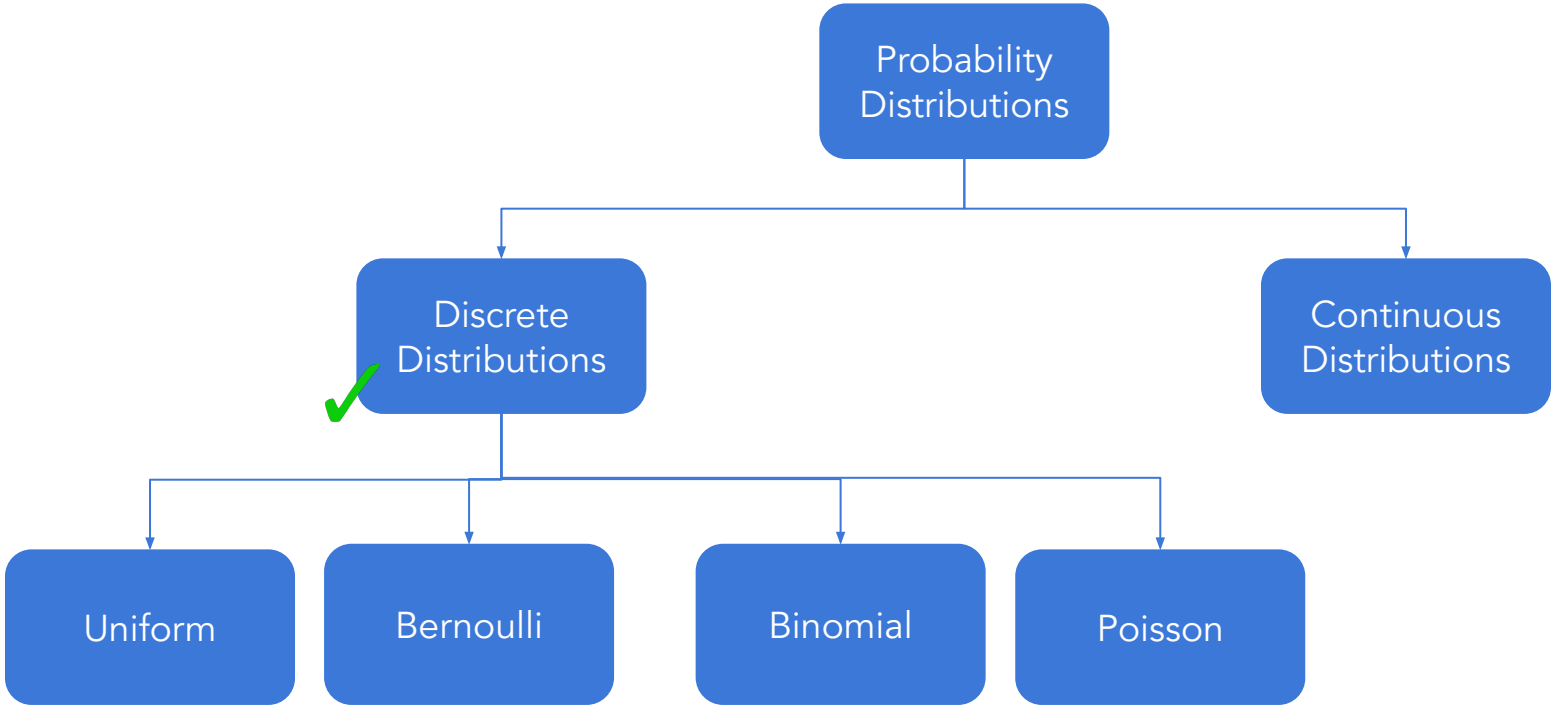
- It is the mathematical mean of squares of deviations taken from the mean
- Denoted by σ^2
- For a probability distribution, it is $\sigma^2 = \sum (x - \mu)^2 . f(x)$

  where f(x) is the probability mass function

Probability distributions



Probability distributions



Discrete probability distribution

- A discrete distribution comprises of the probabilities of the outcomes of a random variable with finite values and is used to model a discrete random variable

- A discrete probability distribution is defined by its probability mass function (p.m.f)

- The mathematical expectation E(X) and variance V(X) of a discrete probability distribution is given by

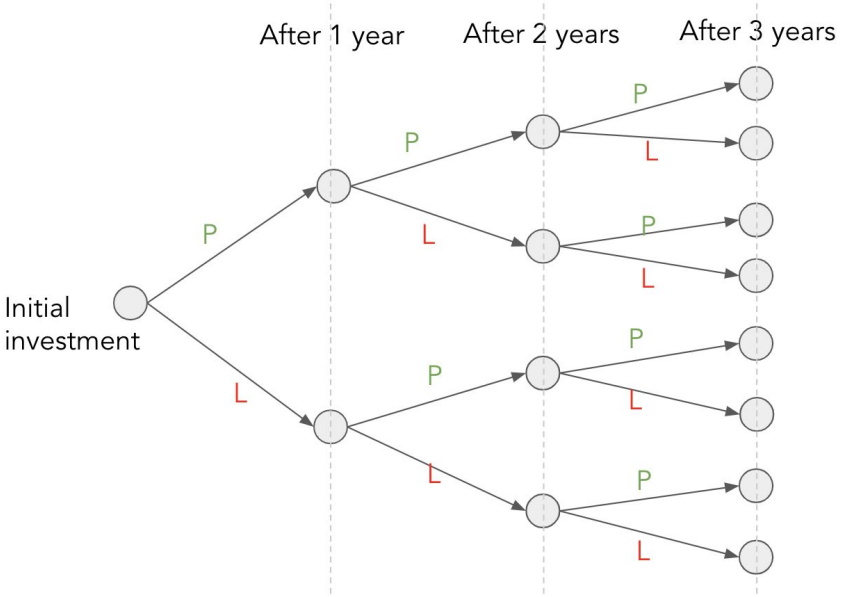
$$E(X) = \sum_i x_i \cdot P(X = x_i)$$

$$Var(X) = E[(X - \mu)^2] = \sum_i (x_i - \mu)^2 \cdot P(X = x_i)$$

(Refer appendix A.1)

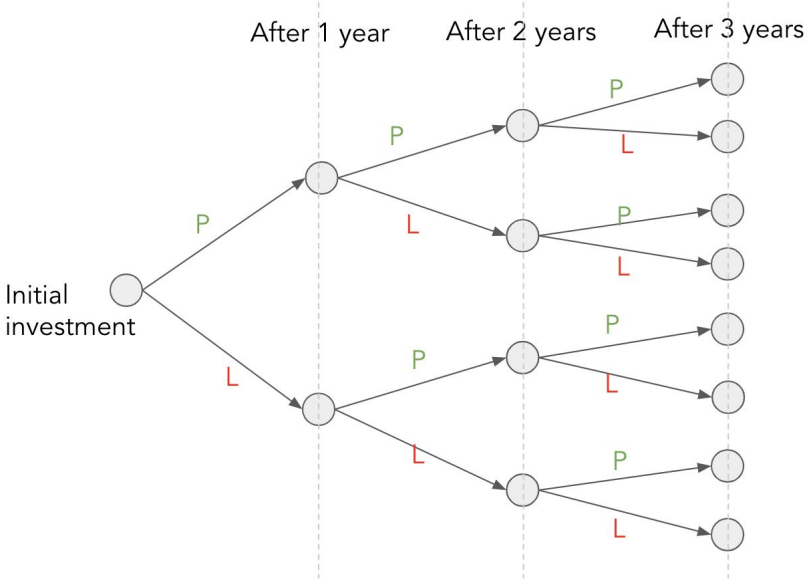
Binomial distribution

Steve has invested in a scheme. The following tree shows all the possibilities of how his investments may vary year to year over a period of three years



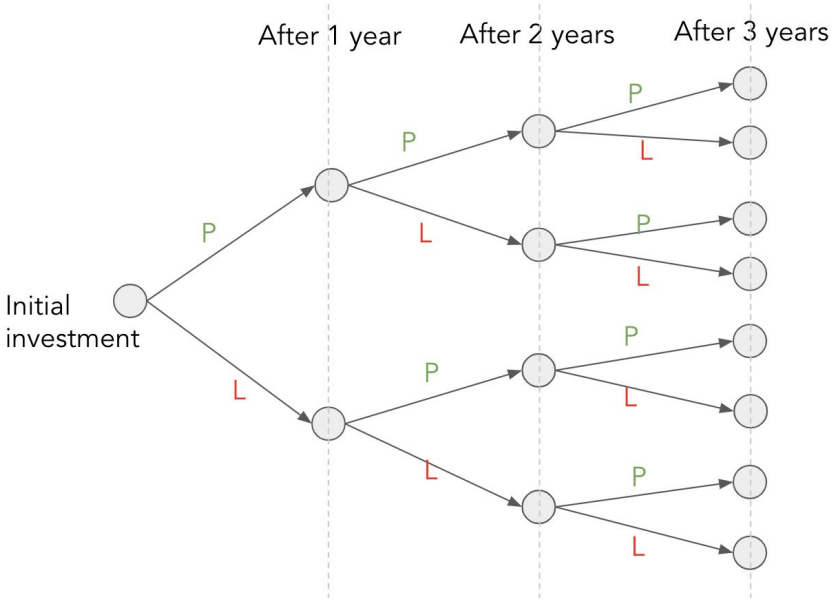
Binomial distribution

- Let X be the event of a change in the initial invested amount
- There is a profit with probability *p*, naturally the loss incurred will have the probability *1-p = q* (say)
- Let the probability of profit be 0.4 i.e. P(X = P) = p = 0.4
- Let the probability of loss be 0.6 i.e. P(X = L) = 1 - p = 1- 0.4 = 0.6



Binomial distribution

The probability of each of the possible scenarios is calculated as follows:



Combination	Number of Profits	Probability of the combination
PPP	3	0.4 × 0.4 × 0.4 = 0.064
PPL	2	0.4 × 0.4 × 0.6 = 0.096
PLP	2	0.4 × 0.6 × 0.4 = 0.096
PLL	1	0.4 × 0.6 × 0.6 = 0.144
LPP	2	0.6 × 0.4 × 0.4 = 0.096
LPL	1	0.6 × 0.4 × 0.6 = 0.144
LLP	1	0.6 × 0.6 × 0.4 = 0.144
LLL	0	0.6 × 0.6 × 0.6 = 0.216
Total probability		1

Binomial distribution

We can easily tabulate these probabilities.

Combination	Number of Profits	Probability of the combination
PPP	3	$0.4 \times 0.4 \times 0.4 = 0.064$
PPL	2	$0.4 \times 0.4 \times 0.6 = 0.096$
PLP	2	$0.4 \times 0.6 \times 0.4 = 0.096$
PLL	1	$0.4 \times 0.6 \times 0.6 = 0.144$
LPP	2	$0.6 \times 0.4 \times 0.4 = 0.096$
LPL	1	$0.6 \times 0.4 \times 0.6 = 0.144$
LLP	1	$0.6 \times 0.6 \times 0.4 = 0.144$
LLL	0	$0.6 \times 0.6 \times 0.6 = 0.216$
Total probability		1

Number of Profits	Probability
0	0.216
1	0.144
2	0.096
3	0.064

Binomial distribution

- It is based on the Bernoulli distribution
- A discrete random variable (X) taking values 0,1,2,...,n follows a binomial distribution if the p.m.f. of X is given as:

$$P(x) = \binom{n}{x} p^x q^{n-x} \qquad x = 0, 1, 2, \ldots, n$$
$$= 0 \qquad otherwise$$

- X follows binomial distribution with parameters *n* and *p*, i.e X ~ Binomial(n,p)
- *p* is the probability of success, *q* is the probability of failure and *n* is the number of times the experiment is conducted

Binomial distribution

Recall Steve's example, multiply each probability with ${}^{n}C_{x}$

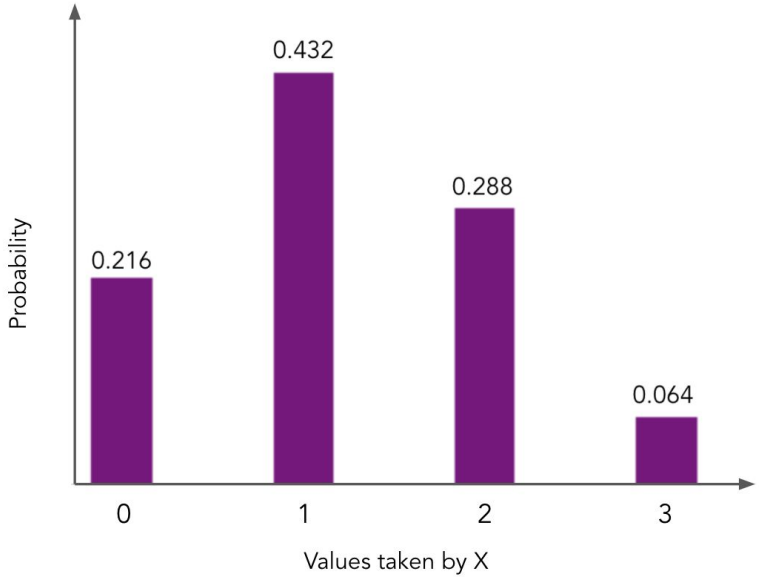
Number of Profits (x)	Probability	${}^{3}C_{x}$	P(x) = ${}^{3}C_{x}$ Probability
0	0.216	${}^{3}C_{0} = 1$	0.216
1	0.144	${}^{3}C_{1} = 3$	0.432
2	0.096	${}^{3}C_{2} = 3$	0.288
3	0.064	${}^{3}C_{3} = 1$	0.064

$$P(x) = \binom{n}{x} p^x q^{n-x} \qquad x = 0, 1, 2, \ldots, n$$
$$= 0 \qquad otherwise$$

Binomial distribution

The probability of the Binomial(3, 0.4)

Number of Profits (x)	$P(x) = {}^{3}C_{x}$ Probability
0	0.216
1	0.432
2	0.288
3	0.064



If *n* independent random variables are Bernoulli distributed, then the sum of these random variables follows a Binomial distribution

Binomial distribution

The mean and the variance of the distribution is given as:

$$Mean = E(X) = \sum_{x=0}^{n} xP(x)$$

$$= \sum_{x=0}^{n} x\binom{n}{x}p^x q^{n-x}$$

$$= np$$

$$\text{Variance} = \mathrm{E}(\mathrm{X}^2) - [\mathrm{E}(\mathrm{X})]^2$$

$$= \mathrm{npq}$$

For our example, if n = 10, p = ⅔ and q = ⅓ . Then, E(X) = 20/3 and V(X) = 20/9

Binomial distribution

Question:

In a shooting academy, data was collected on the precision shooting of a student. From 15 shots fired 11 were on target. Considering the same student, what is the probability that out of 50 shots fired, exactly 35 will hit the target?

Binomial distribution

Solution:

Let X denote the shot fired hits the target

The probability of success, i.e. is hitting the target is 11/15
Thus, p = 0.733

To find: the probability that out of 50 shots fired, 35 will hit the target

Here, n = 50

We have X~Binomial(50, 0.733)

Binomial distribution

Solution:

The required probability is P(X = 35)

$P(X = x) = {}^{n}C_{x}\, p^{x}\, (1-p)^{n-x}$

$P(X = 35) = {}^{50}C_{35}\, (0.733)^{35}\, (1-0.733)^{50-35}$

$= {}^{50}C_{35}\, (0.733)^{35}\, (0.267)^{15}$

$= 0.107$

Thus, the probability that out of 50 shots fired, exactly 35 will hit the target is 0.107

Poisson distribution

- It is used to determine the probability of events that occur in a fixed interval of time/space

- It is used to describe the distribution of rare events in a large population

- A discrete random variable X follows a Poisson distribution with parameter λ, if p.m.f. of X is given as:

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x\,!} \qquad x = 0, 1, 2, \ldots \quad \lambda > 0$$

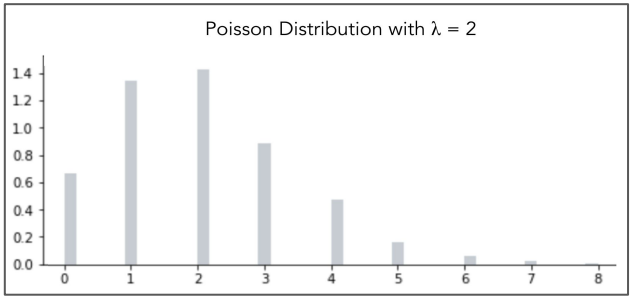
Where,

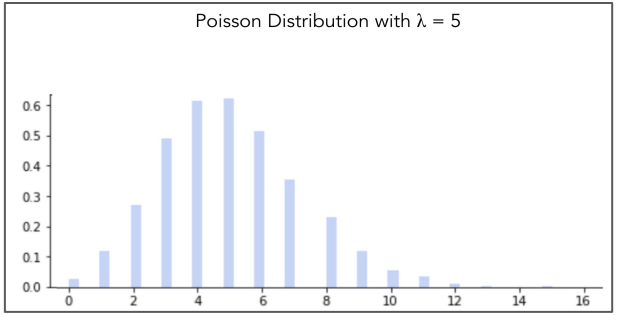
e = Euler's number (= 2.718)

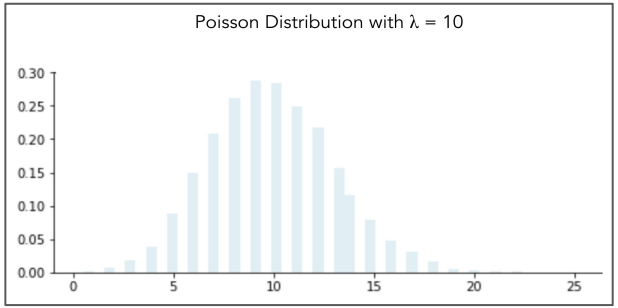
x! = Factorial of x

Poisson distribution

The Poisson distribution is positively skewed.







Poisson distribution

The mean and the variance of the distribution is given as:

$$Mean = E(X) = \sum_{x=0}^{\infty} xP(x)$$
$$= \lambda$$

$$\text{Variance} = \mathrm{E}(\mathrm{X}^2) - [\mathrm{E}(\mathrm{X})]^2$$
$$= \lambda$$

Mean and variance of the poisson distribution are equal and it is the parameter of distribution.

WANT TO KNOW MORE?

Poisson as a limiting case of binomial distribution

Let X~Binomial(n, p) such that for large n ($n \to \infty$) and small p, $np = m$ is a constant then X approaches Poisson(m)

Poisson distribution

Question:

A call center receives an average of 15 calls per hour. The arrival of calls is random, and the distribution of the number of calls received per hour can be modeled as a Poisson process.

1. What is the probability that the call center will receive exactly 20 calls in a given hour?

Poisson distribution

Solution:

Let X: number of calls received by call center

The average number of calls per hour is **15**, so the Poisson parameter for this problem is: λ=15 calls/hour

We know P(X=x) = f(λ)

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Poisson distribution

Solution:

Probability of Receiving Exactly 20 Calls in an Hour

The probability mass function (PMF) of a Poisson distribution is given by:

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

We have,

$$P(X = 20) = \frac{15^{20}e^{-15}}{20!}$$

Poisson distribution

Solution:

We can calculate the components individually:

$$P(X = 20) = \frac{15^{20} \cdot e^{-15}}{20!}$$

$$P(X = 20) = \frac{3.27 \times 10^{23} \cdot 3.06 \times 10^{-7}}{2.43 \times 10^{18}}$$

$$P(X = 20) = \frac{1.00 \times 10^{17}}{2.43 \times 10^{18}} \approx 0.0412$$

The probability that the call center will receive **exactly 20 calls in a given hour** is approximately **0.0412**, or **4.12%**

A local bakery receives an average of 12 customer orders per hour. The owner wants to determine the likelihood of receiving exactly 10 orders in a particular hour, as well as the probability of receiving no more than 5 orders during the same hour. Understanding these probabilities will help the bakery staff manage their inventory and staffing needs more effectively. Additionally, the owner is interested in the overall probability of receiving more than 5 orders during the busiest part of the day, which lasts for 3 hours.

Soln:

- A local bakery receives an average of 12 customer orders per hour.
- The owner wants to determine:
  1. Probability of receiving exactly 10 orders in an hour.
  2. Probability of receiving no more than 5 orders in an hour.
  3. Probability of receiving more than 5 orders during a 3-hour busy period.

$\lambda = 12$

$e \approx 2.71828$

1 - Probability of Exactly 10 Orders

Calculation for $P(X=10)$:

$$P(X = 10) = \frac{e^{-12} \cdot 12^{10}}{10!}$$

- **Detailed Calculation:**
  - $e^{-12} \approx 6.144212 \times 10^{-6}$
  - $12^{10} = 61917364224$
  - $10! = 3628800$

$$P(X = 10) = \frac{(6.144212 \times 10^{-6}) \cdot 61917364224}{3628800} \approx 0.0001045$$

2 - Probability of Receiving No More Than 5 Orders

- **Calculation for $P(X \leq 5)$:**
- Sum of probabilities up to 5:
  - $P(X = 0) = e^{-12}$
  - $P(X = 1) = \frac{12^1 \cdot e^{-12}}{1!}$
  - $P(X = 2) = \frac{12^2 \cdot e^{-12}}{2!}$
  - $P(X = 3) = \frac{12^3 \cdot e^{-12}}{3!}$
  - $P(X = 4) = \frac{12^4 \cdot e^{-12}}{4!}$
  - $P(X = 5) = \frac{12^5 \cdot e^{-12}}{5!}$
- **Results:**
  - $P(X = 0) \approx 0.000006144$
  - $P(X = 1) \approx 0.00007374$
  - $P(X = 2) \approx 0.0004351$
  - $P(X = 3) \approx 0.0008458$
  - $P(X = 4) \approx 0.0016944$
  - $P(X = 5) \approx 0.0040917$

$$P(X \leq 5) \approx 0.0064184$$

3 - Probability of More Than 5 Orders in 3 Hours

- Average customer orders for 3 hours (λ): $12 \times 3 = 36$
  - **Calculate for $\lambda = 36$:**
  - Calculate $P(X \leq 5)$ for $k = 0, 1, 2, 3, 4, 5$

1. **Overall Calculations for 3 hours:**
   - $P(X = 0) = e^{-36} \approx 1.042 \times 10^{-16}$
   - $P(X = 1) = \frac{36^1 \cdot e^{-36}}{1!} \approx 3.750 \times 10^{-15}$
   - $P(X = 2) = \frac{36^2 \cdot e^{-36}}{2!} \approx 6.749 \times 10^{-14}$
   - $P(X = 3) = \frac{36^3 \cdot e^{-36}}{3!} \approx 8.099 \times 10^{-13}$
   - $P(X = 4) = \frac{36^4 \cdot e^{-36}}{4!} \approx 7.299 \times 10^{-12}$
   - $P(X = 5) = \frac{36^5 \cdot e^{-36}}{5!} \approx 5.016 \times 10^{-11}$

$$P(X \leq 5) \approx 5.860 \times 10^{-11}$$

- **Final Calculation:**

$$P(X > 5) = 1 - P(X \leq 5) \approx 1 - 5.860 \times 10^{-11} \approx 0.9999999999414$$

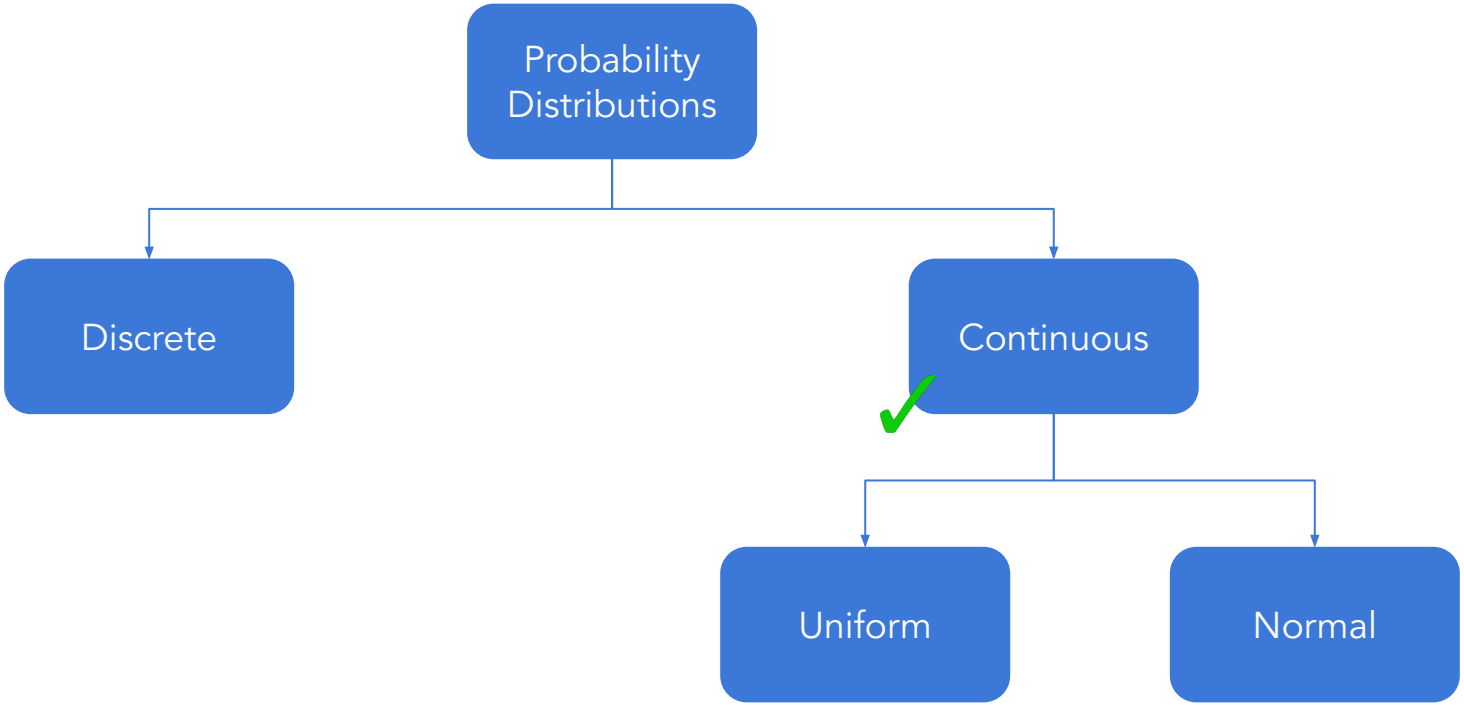
Calculations Result:

- Probability of receiving exactly 10 orders in an hour: 0.0001045
- Probability of receiving no more than 5 orders in an hour: 0.0064184
- Probability of receiving more than 5 orders during a 3-hour period: ≈ 1 (very high likelihood)

Summary

- Some of the discrete distributions are: Uniform, Bernoulli, Binomial and Poisson
- Discrete distribution is determined using probability mass function (p.m.f.)
- In uniform distribution, all the values of the variable are equally likely
- The sum of independent and identically distributed Bernoulli variables follows binomial distribution
- Poisson distribution can be considered as a limiting case of a binomial distribution

Probability distributions



Continuous probability distribution

- It is a probability distribution of the continuous random variable
- The continuous distributions are defined by their probability density function (p.d.f.) denoted as f(x)
- The probability is given by the area under the pdf on a specified range. Thus, the value of the probability density function is 0 at a particular point
- The area under the curve is always equal to 1. Therefore, is considered as a probability distribution

Normal distribution

The probability density function for the normal distribution is,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

It is said that X ~ N(μ, σ^2), i.e the variable X follows the normal distribution with parameter μ and σ^2

Normal distribution

The mean and the variance of the distribution is given as:

$$Mean = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

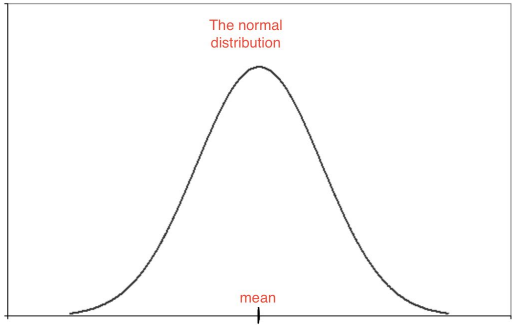
$$= \mu$$

$$V(X) = E(X) - [E(X)]^2$$

$$= \sigma^2$$

Normal distribution

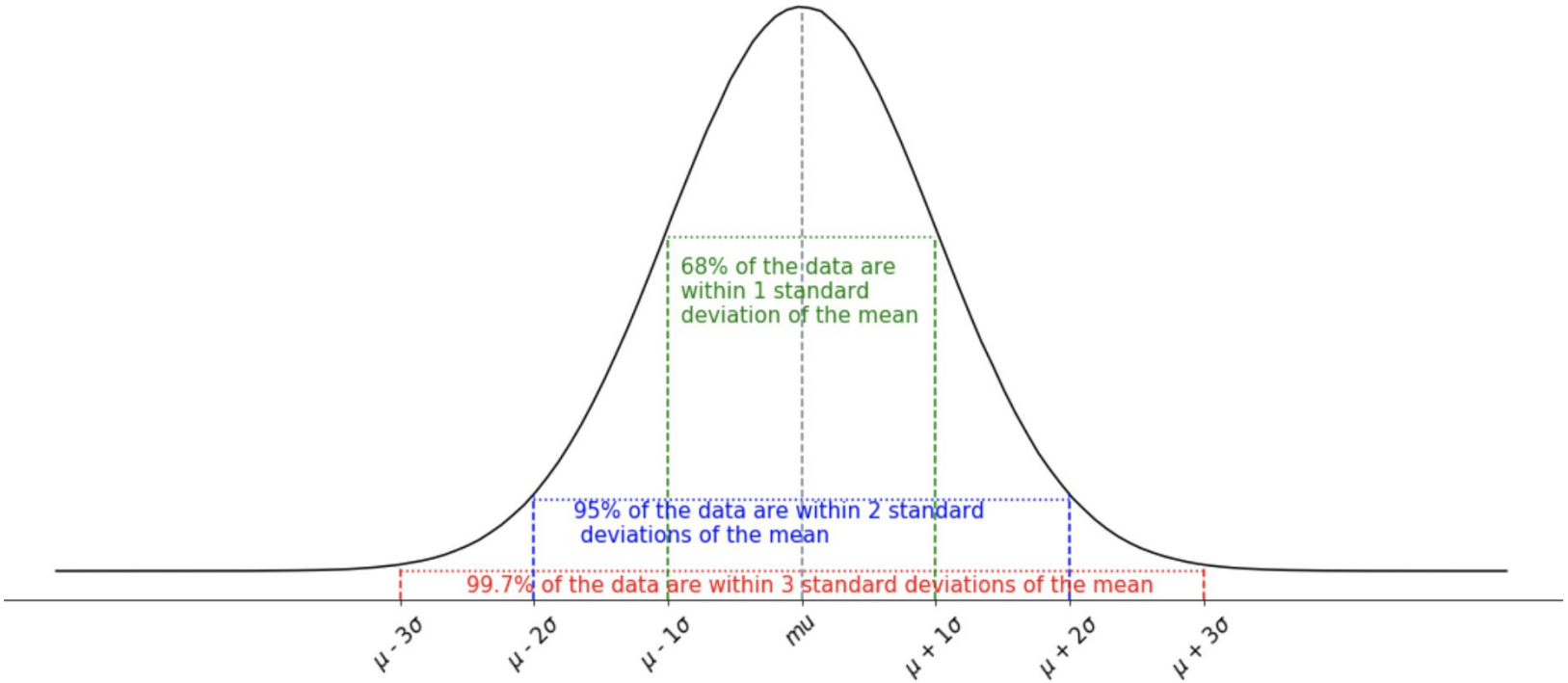
- Also called as the Gaussian distribution
- The frequent observations are found in the middle of the distribution, and further decrease away towards the tails
- A normal distribution has a bell-shaped density curve described by its mean and standard deviation
- For normally distributed data,
  - Mean = Mode = Median
  - Symmetric about the center



Characteristics of normal distribution

- The distribution is symmetric about the mean
- For $X \sim N(\mu, \sigma^2)$, $P(X > a) = P(X < -a)$ due to symmetricity
- The standard normal distribution has mean 0 and variance 1
- For $X \sim N(\mu, \sigma^2)$, if $Z = (x - \mu)/\sigma$ then $Z \sim N(0, 1)$ that is the standard normal distribution

The normal distribution spread

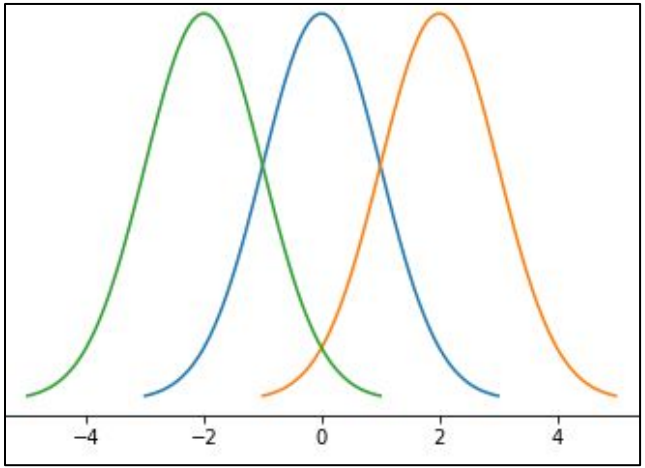


Normal distribution

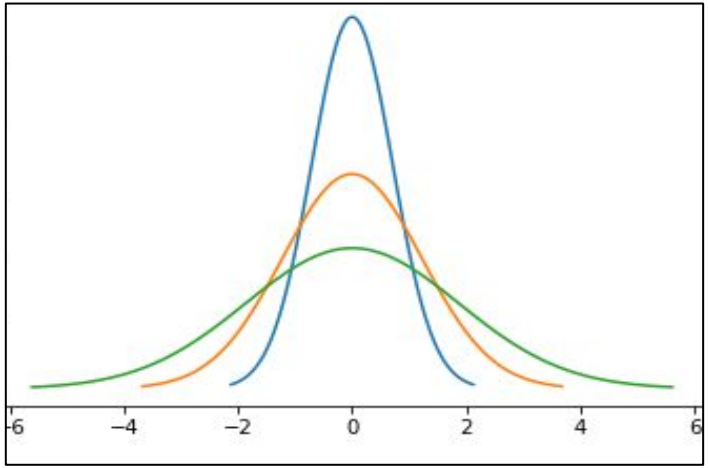
- IQ of the human population is the example of normal distribution
- Most of the people in the specific population have average IQ
- The number of people with higher IQ and lower IQ than the average IQ is almost equal, and a very less number of people have extremely less IQ or extremely high IQ

Different shapes of normal distribution

Normal distributions with same standard deviation but different means:



Normal distributions with same mean but different standard deviations:



Normal distribution

Question:

A monthly balance in the bank account of credit card holders is assumed to be normally distributed with mean $500 and variance $100. What is the probability that the balance can be more than $513.5?

Normal distribution

Solution:

Let X: monthly balance in the bank account of a credit card holders

$X \sim N(\mu = 500, \sigma^2 = 100)$

To find: the probability that the balance can be more than $513.5

The required probability is $P(X > 513.5)$

Normal distribution

Solution:

$$P(X > 513.5) = \int_{513.5}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{\sigma^2}} dx$$

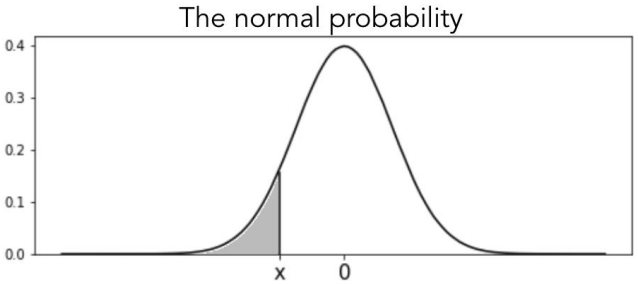
$$= \int_{513.5}^{\infty} \frac{1}{\sqrt{2\pi \times 100}} e^{-\frac{(x-500)^2}{100}} dx$$

$$= \int_{513.5}^{\infty} \frac{1}{25.07} e^{-\frac{(x-500)^2}{100}} dx$$

… Here is a problem. We see it is difficult to solve the integral, so we make use of the normal tables.

The normal table

- Let the r.v. X strictly follow the standard normal distribution
- Let r.v. X take a negative value x
- This table gives the lower tail probabilities; $P(X \leq x)$

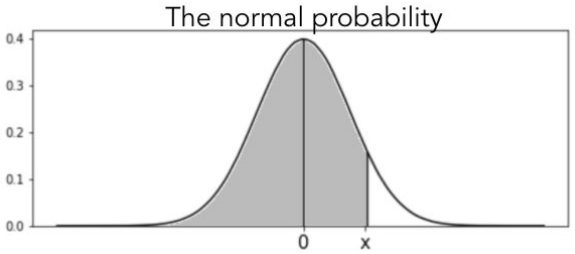


z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
−3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Table Ref: Link

The normal table

- Let the r.v. X strictly follow the standard normal distribution
- Let r.v. X take a positive value x
- This table gives the lower tail probabilities; $P(X \leq x)$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

Table Ref: Link

Normal distribution

Solution:

We know that for $X \sim N(\mu, \sigma^2)$, if $Z = (x - \mu)/\sigma$ then $Z \sim N(0, 1)$ that is the standard normal distribution

$X \sim N(\mu = 500, \sigma^2 = 100)$ thus $Z = (x - 500)/\sqrt{100} \sim N(0, 1)$

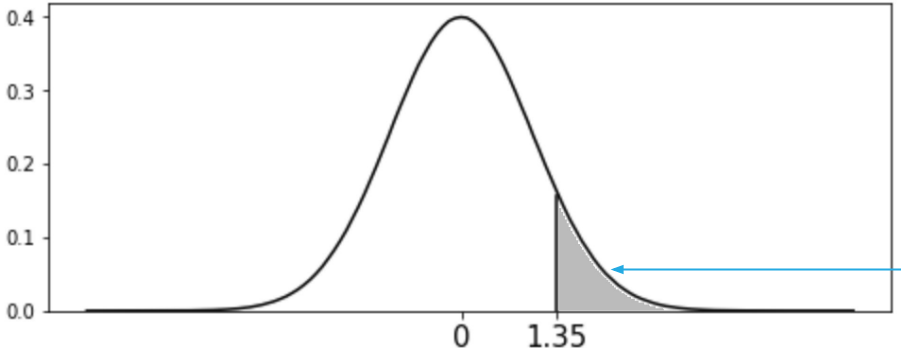
The required probability is $P(X > 513.5)$, now becomes

$$P(X > 513.5) = P(\frac{X-\mu}{\sigma} > \frac{513.5-500}{\sqrt{100}}) = P(Z > \frac{513.5-500}{\sqrt{100}}) = P(Z > 1.35)$$

Normal distribution

Solution:

Thus the required probability now is P(Z > 1.35)



The shaded region represents the required probability

Now we refer to the normal table to the probability

Normal distribution

Solution:

The table gives P(Z ≤ 1.35).

1. Look horizontally in the row of 1.3
2. Consider the value corresponding column to 0.05

2

1

z	0.00	0.01	0.02	0.03	0.04	0.05	0.0
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.52
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.56
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.60
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.64
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.67
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.71
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.74
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.77
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.80
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.83
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.85
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.87
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.89
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.91
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.92
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.94

Normal distribution

Solution:

Since we get the lower tail probability as 0.9115

The required probability $P(Z > 1.35) = 1 - P(Z \leq 1.35)$

$= 1 - 0.9115$

$= 0.0885$

The probability that the balance can be more than \$513.5 is 0.0885

Question:

The heights of students in a school are normally distributed with a mean (μ) of 65 inches and a standard deviation (σ) of 4 inches.

Questions:

1. What percentage of students are shorter than 60 inches?
2. What is the height below which 90% of the students fall?

do it YOURSELF

SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

Solution: 1. What percentage of students are shorter than 60 inches?

- We want to find $P(X < 60)$.
- We first calculate the z-score:

$$z = \frac{X - \mu}{\sigma} = \frac{60 - 65}{4} = -1.25$$

- Using the z-table or a calculator, $P(Z < -1.25) \approx 0.1056$.
- This means approximately **10.56%** of students are shorter than 60 inches.

do it YOURSELF

SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

Solution: 2. What is the height below which 90% of the students fall?

- We need to find the height corresponding to the 90th percentile (i.e., $P(X < x) = 0.90$).
- The z-score for 90% is approximately $z \approx 1.28$.
- Using the z-score formula:

$$X = \mu + z\sigma = 65 + 1.28(4) \approx 69.12$$

- Therefore, **90% of the students are below approximately 69.12 inches.**

Summary

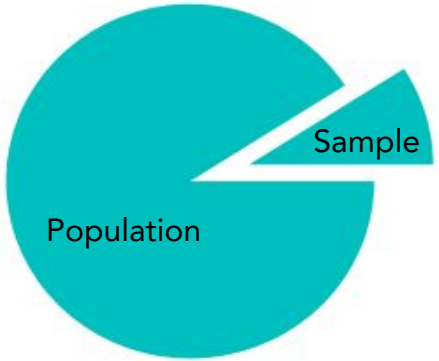
- Continuous distributions: Uniform and normal
- Continuous distribution is determined using probability density function (p.d.f.)
- The value of pdf at a particular point is always zero
- For a normal distribution, Mean = Median = Mode
- Standard normal distribution has mean zero and variance one

Population and Sample

Population & Sample

- Population is a collection of all the individuals or objects

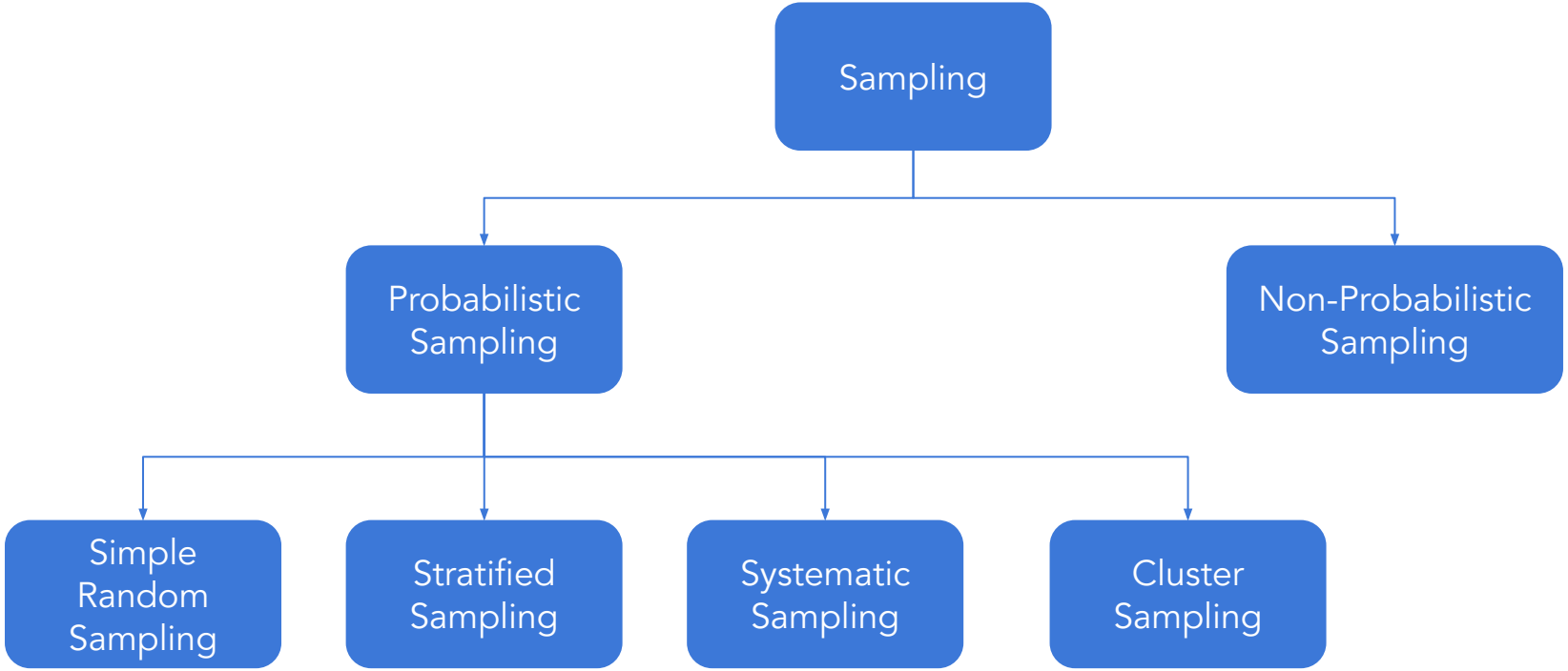
- Sample is a subset of the population which is a representative of the population



Need to draw a sample

- In some of the real case scenarios we may not be able to gather all the observations in the population
- In such cases we can consider a sample which can be used to draw inferences about the population
- Sampling can reduce the computational time and cost
- In machine learning models, a sample obtained using an appropriate sampling strategy can provide accurate results with the less number of observations

Sampling techniques



Simple random sampling (SRS)

- Simplest and most common method
- Sample is drawn unit by unit
- Each unit has equal probability of being selected
- Two ways of selecting units:
  - Without replacement: if the selected unit is not returned to the population
  - With replacement: if the selected unit is returned to the population

SRS without replacement

- At any specific stage in the sampling, the probability of any of the remaining units is 1/N, where N is the total number of sample units
- From N sample units, if a sample of size n is drawn, there are ${}^{N}C_{n}$ possible samples
- The probability of selecting one of them is $1/{}^{N}C_{n}$

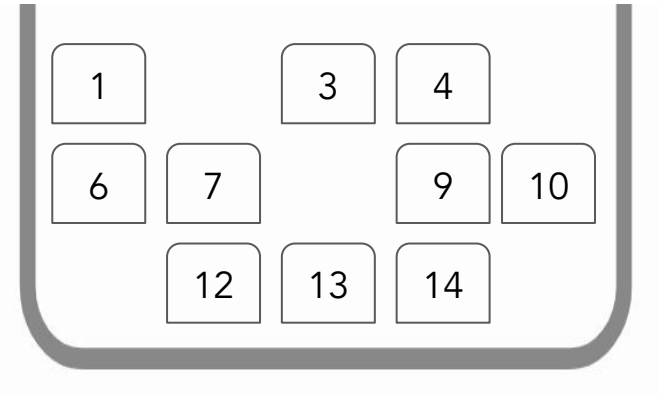
SRS without replacement

The unit selected to be in the sample is not returned to the population.

SRS without replacement - example

Drawing a sample from manufactured articles to know the proportion of defectives.

Manufactured articles

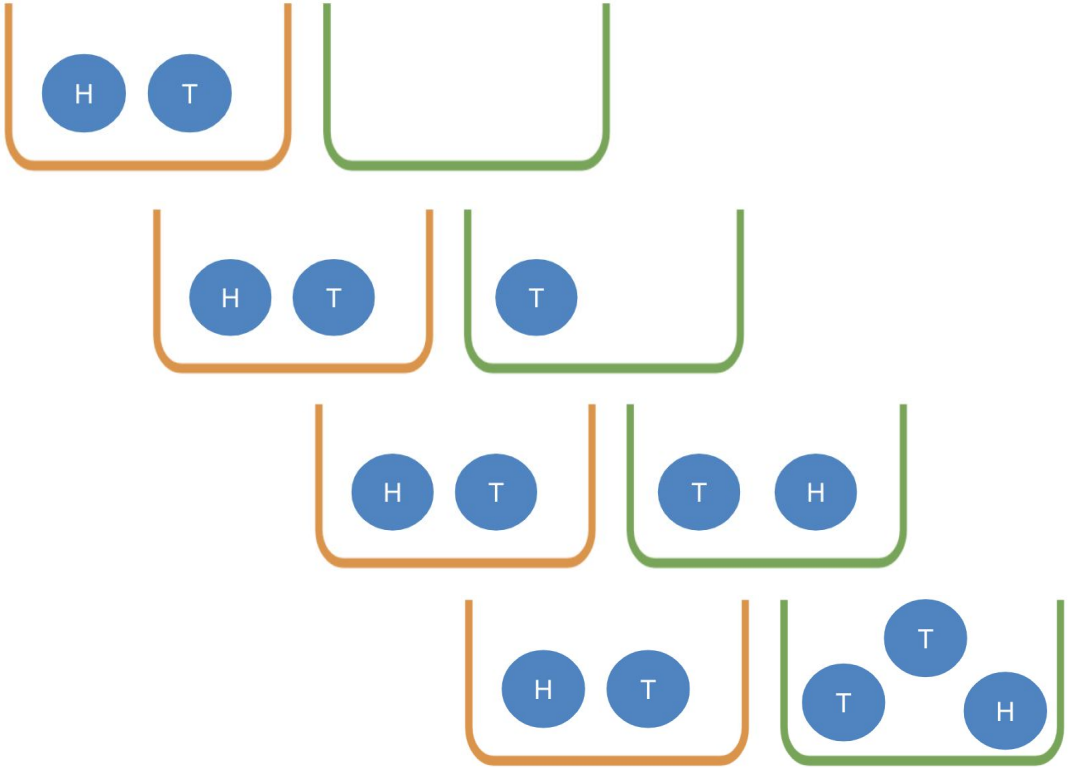


Sample to test for defective articles

SRS with replacement

- The probability of selecting a specific unit at any given draw is always 1/N, where N is total number of sample units
- From N sample units, if sample of size n is drawn, there are N^n possible samples
- The probability of selecting one of them is $1/N^n$
- Example: You have 5 fruits and want to take a sample of 2. You can choose a fruit more than once.

SRS with replacement



The unit selected to be in the sample is returned to the population.

SRS with replacement - example

- Generally used in simulation studies
- A car insurance company wants to set the premium amount. The probabilities for major and minor accident are known to be 0.02% and 2% respectively
  (assume that an insuree can meet with an accident only once)
- A SRS with replacement is conducted to simulate 1 million such scenarios

Summary

Sampling Method	Summary
Simple Random	Mostly used method. The sample can be selected with or without replacement.
Stratified	Sample point is selected from each subgroup in the population.
Systematic	Easy to implement for large population. The first sample point is selected randomly and remaining points are selected at fixed intervals.
Cluster	An entire subgroup in the population is selected as a sample point.

Summary

- Sample is the subset of the population
- Different sampling methods: SRS with replacement (SRSWR), SRS without replacement (SRSWOR), Stratified, Systematic, Cluster sampling
- SRSWR method can be used if one wish to select the sample observation more than once
- Stratified sampling can be used when data is heterogeneous
- Cluster sampling can be used when the data consists of several groups

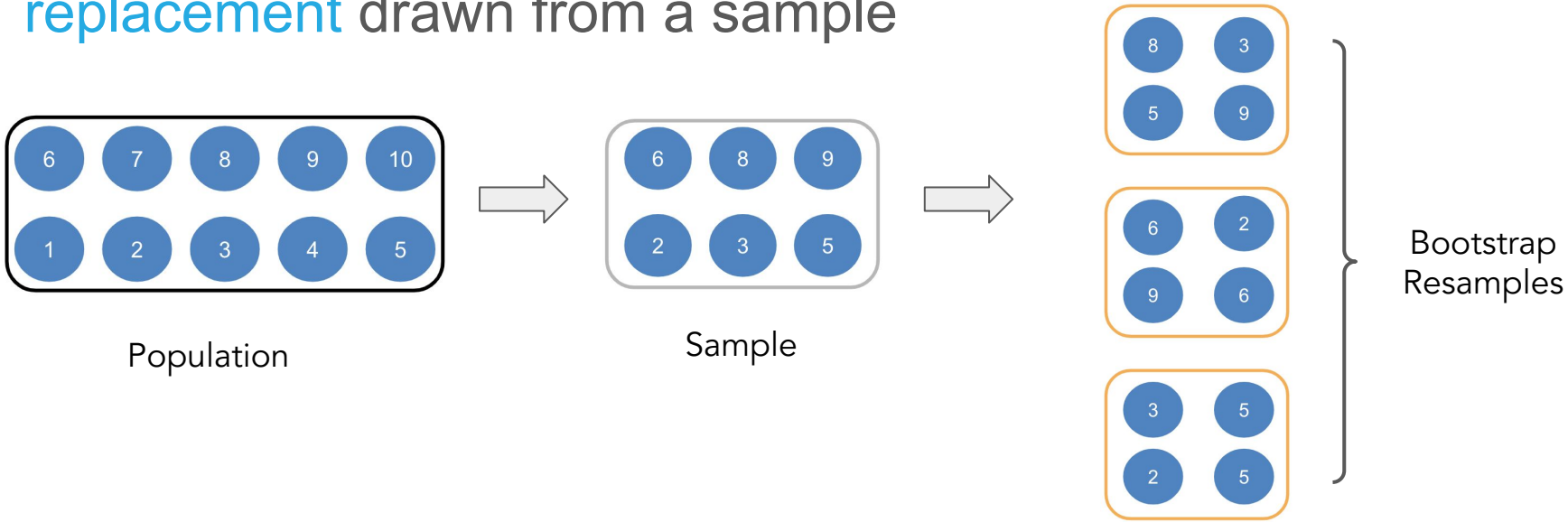
Resampling

Resampling

- Resampling is a technique in which we draw a sample from the available sample
- Resampling often involves performing similar testing methods with sample sizes within that group.
- This can mean testing the same sample, or reselecting samples that can provide more information about a population.
- The resampling techniques are:
  - Bootstrap resampling
  - Jackknife resampling

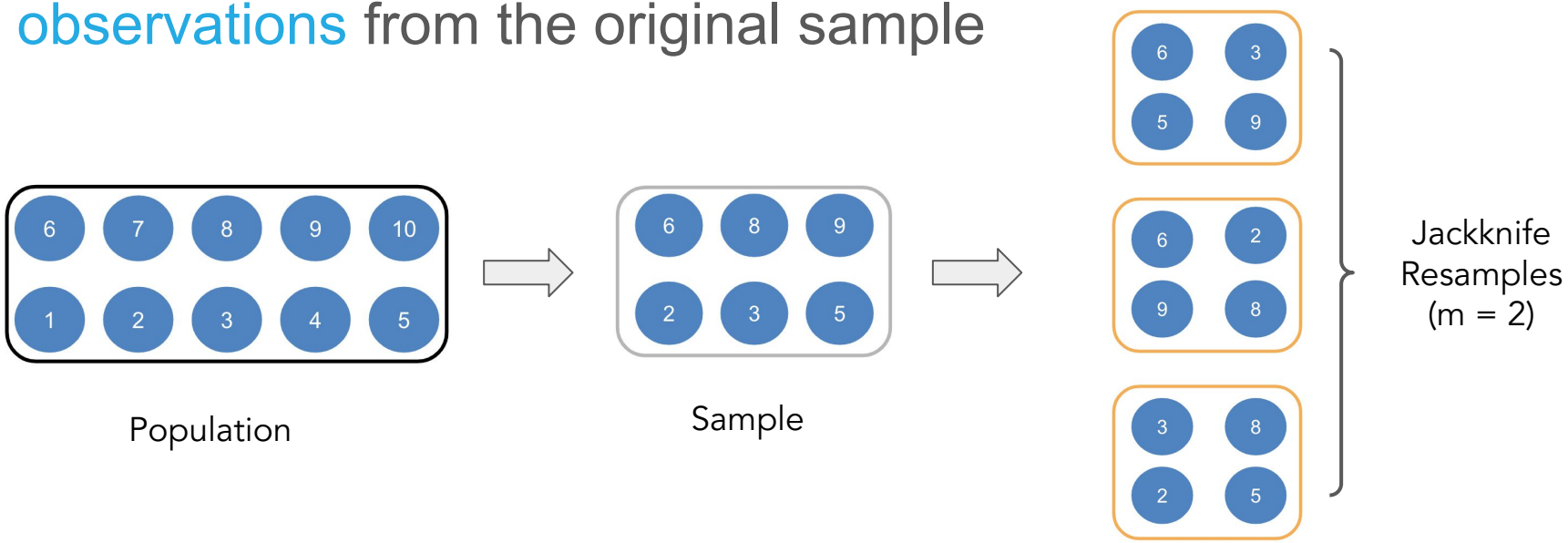
Bootstrap resample

Bootstrap resample is a simple random sample with replacement drawn from a sample



Jackknife resample

Jackknife resample is a method in which one deletes m observations from the original sample



A machine is used for packing grains in a sack. The weight of every bag is recorded as follows:

15.6	15.7	14.6	14.7	15.8	15.0	15.1	14.9
14.8	15.5	15.4	15.3	14.8	15.1	14.8	14.9

Draw at least the two samples and obtain their means. Compare this with the population mean.

Sampling variability

- Let us draw four different samples from the data
- The sample mean varies from sample to sample
- This intuitive idea is called sampling variability
- The sample mean is close to the population mean

Population Mean
15.125

Sample	Sample Mean
14.7, 15.1, 15.0, 15.6, 14.9, 15.4, 15.8, 15.1	15.20
15.8, 15.5, 15.3, 15.4, 15.8, 14.8, 14.6, 14.8	15.250
15.7, 15.6, 14.8, 14.8, 14.7,14.9, 15.6, 15.1	15.150
14.8, 15.1, 15.7, 14.6, 15.6, 15.4, 15.6, 15.0	15.225

Parameter and Statistic

- Parameter:
  - The population quantities are termed as a parameter
  - In inferential statistics, sample is used to estimate the population parameters
  - Example: Population mean

- Statistic:
  - Statistical measure computed from sample observations
  - Also known as an estimator
  - Example: Sample mean

Parameter and Statistic

The statistical notations for population parameters and sample statistics

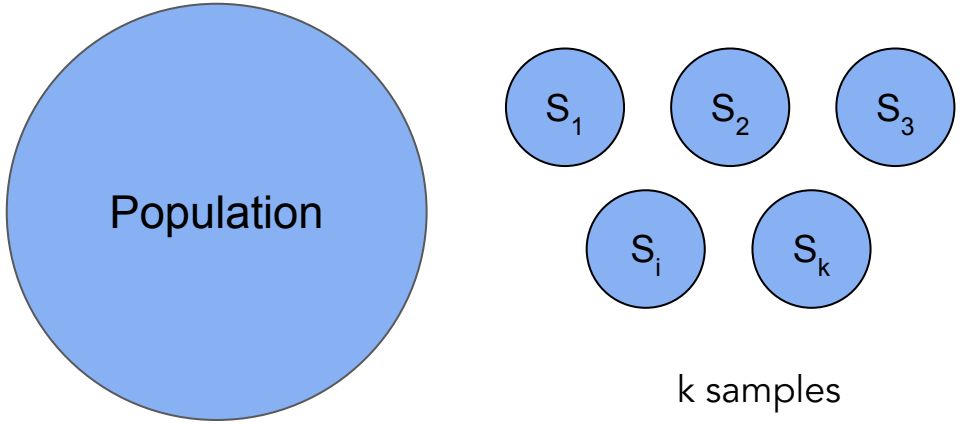
	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\overline{p}$

Sampling Distributions

Sampling distribution

- A total of ${}^{N}C_{n}$ (= k say) samples can be drawn from a population of size N and sample size n

- Obtain the mean and variance of each of these samples



k samples

Sampling distribution of sample mean

- The set of random sample means $\bar{x}_1$, $\bar{x}_2$,..., $\bar{x}_k$ are the observations for sampling distribution of sample mean

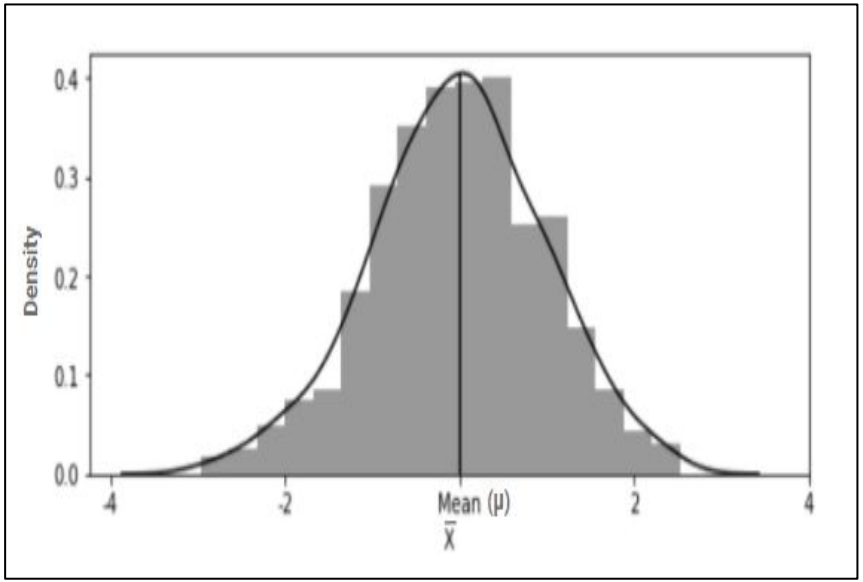
- These sample means follow the normal distribution

- Also, the mean of sample means is the population mean $\mu_{\bar{x}} = \mu$

Sample No	Sample Mean
1	$\bar{x}_1$
2	$\bar{x}_2$
3	$\bar{x}_3$
⋮	⋮
k	$\bar{x}_k$

Sampling distribution of sample mean

- The mean of sampling distribution of sample mean is $\mu_{\bar{x}}$



Central limit theorem

- The central limit theorem says that the sampling distribution of the mean will always be normally distributed, as long as the sample size is large enough.

- Regardless of whether the population has a normal, Poisson, binomial, or any other distribution, the sampling distribution of the mean will be normal.

Central limit theorem

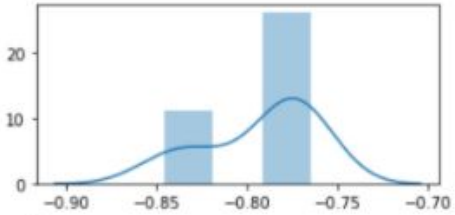
The central limit states that for random samples of large size n (usually n > 30) taken from a single population with mean μ and standard deviation σ, the sampling distribution of mean follows a normal distribution with mean $\mu_{\bar{x}}$ and standard deviation σ/√n or s/√n which ever is known.

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0, 1)$$

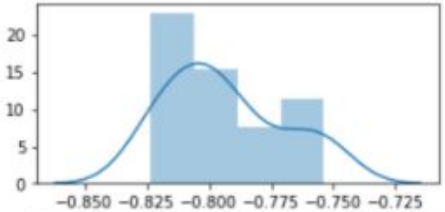
PLEASE!! NOTE

Central limit theorem

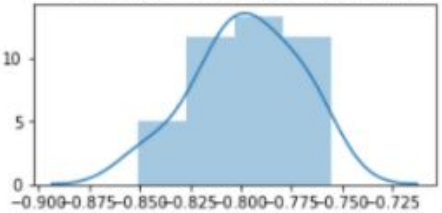
nSamples: 10



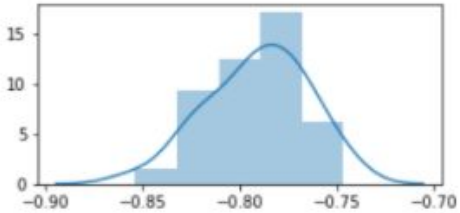
nSamples: 15



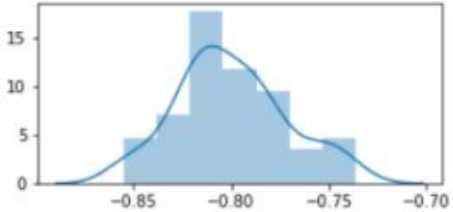
nSamples: 25



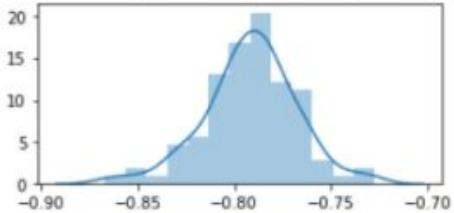
nSamples: 30



nSamples: 50



nSamples: 100



do it YOURSELF

A coffee shop claims that the average time customers spend in the shop is 10 minutes, with a standard deviation of 2 minutes. You take a random sample of 50 customers. What is the probability that the average time spent by these customers is greater than 10.5 minutes?

Solution:

Population mean (μ) = 10 minutes

Population standard deviation (σ) = 2 minutes

Sample size (n) = 50 customers

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{50}} \approx \frac{2}{7.07} \approx 0.283$$

$$z = \frac{X - \mu}{SE} = \frac{10.5 - 10}{0.283} \approx \frac{0.5}{0.283} \approx 1.77$$

$$P(Z < 1.77) \approx 0.9616$$

To find $P(Z > 1.77)$:

$$P(Z > 1.77) = 1 - P(Z < 1.77) = 1 - 0.9616 \approx 0.0384$$

Conclusion:

The probability that the average time spent by the 50 customers is greater than 10.5 minutes is approximately 0.0384, or 3.84%.

Thank you!

Happy Learning :)